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Date: 23/7/26

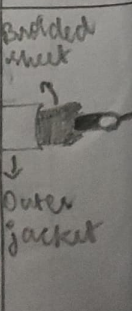
Study of Different types of Network Cables

Aim: Study of different types of cable.

a) Understand different types of network cable

Different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application use	Image
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter **Disadvantages** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet	
	Category 5	upto 100 mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1 Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6a, 6	10 Gbps	**Advantages** • Shielded • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Execative installation effort	Gigabit Ethernet, 10G Ethernet (55m) Usually used in data centers	
SSTP	Category 7	10 Gbps		10Gbit Ethernet, 10G Ethernet (100m)	
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	**Advantages** • High bandwidth • Immune to interference • Low loss bandwidth • Versatile	speed of signal is 500m Television network High speed internet connections	

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application use	Image
			Disadvantages • limited distance • cost, size & bulky		
Fibre Optics Cable	Single mode Multi mode	100 Gbps	Advantages • High speed • High bandwidth • high security • Long distance Disadvantages • Expensive • Requires skilled installers.	Maximum distance of fibre optics cable is around 100 meters	

Result: The Types of network cables and making cross wired cables & Straight through cable using clamping / crimping tool.